SEMESTER THIRD

B.TECH

END SEMESTER EXAMINATION

Nov-Dec.-2018

HU-201 ENGINEERING ECONOMICS

Time: 03: 00 Hours

Max.Marks: 50

Note: ***Answer any five questions. Assume suitable missing data if any.***

1(a) Break-even analysis is a very important analytical technique used to study the relationships between costs, revenues and profits. Explain 5

1(b) A manufacturer has following option 5

	Purchase from the market (in Rs.)	Manufacture within the company (in Rs.)
Selling price /unit	6000with 18% GST per unit	-
Cost of Machine-1	-	4040440 with 18 % GST
Cost of Machine -2	-	220400 with 12% GST
Cost of raw material/unit		999 with 10% and 08% GST

Annual demand is 1000 units. Suggest to the manufacturer whether he should manufacture or purchase?

2(a) What is the rational for government intervention in the market? In the light of these how would you justify the recent move towards demonetization in the Indian economy? 5

2(b) A transport company has been looking for a new tyre for its truck and has located the following alternatives: 5

Brand	Tyre warranty (months)	Price per tyre (in Rs.)
A	12	1,200
B	24	1,800
C	36	2,100
D	48	2,700

If the company feels that the warranty period is a good estimate of the tyre life and that a nominal interest rate (compounded annually) of 12% is appropriate, which tyre should it buy?

3(a) Discuss the nature of soft drink market in India. What incentive may be given by the Government to encourage Indian producers? 5

3(b) A company must decide whether to buy Machine A or Machine B: 5

	Machine A	Machine B
Initial Cost	Rs. 4,00,000	8,00,000
Useful life, in years	4	4
Salvage value at the end of machine life	Rs. 2,00,000	Rs. 5,50,000
Annual maintenance cost	Rs. 40,000	0

At 12% interest rate, which machine should be selected? (use future worth method of comparison

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4(a) Explain the role of risk, uncertainty and innovation in the determination of profit. 5

4(b) A Company wants to deposit money to create an R&D reserve. The company will get 15,00,000 every year for next 15 years for R&D. The reserve will grow at the rate of 12 per cent annually. Find out the single payment which should be made now. 5

5(a) Environmental problems are macro level problems. Discuss the role of appropriate technology in this regard. Also suggest framework to impose taxes to control environmental hazards. 5

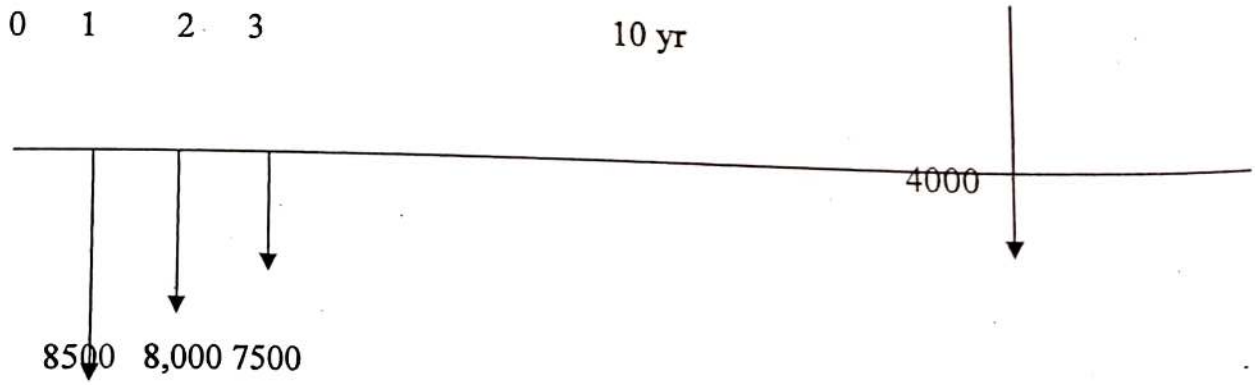
5(b) A firm, Prime Manufacturing is planning to expand its production operation. It has identified three machines which are technologically capable to serve the purpose. The initial outlay and annual revenues with each of the machines are given below: 5

	Initial Outlay (Rs.)	Annual revenue	Life (in years)
Machine I	Rs. 5,45,000	Rs. 2,50,000	15
Machine II	Rs. 6,14,000	Rs. 3,30,000	12
Machine III	Rs. 6,00,900	Rs. 3,50,000	10

If the rate of interest is 12%, which machine the company should opt for? Find out the result by Present Worth method.

6(a) If you are CEO of an MNC, discuss Business Risk which you will consider while deciding location for your foreign venture. 5

6(b) Consider following cash flow diagram. Calculate the total amount at the end of the 10th year at the interest rate of 15%, compounded annually. 5



7(a) Discuss application of Engineering & Technology to improve life at slum. 5

7(b) Discuss salient feature of the Indian Economy 5

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